

12. Bottlenecks Observed

12.1 Introduction

A bottleneck is any component, transaction, or process that limits the performance of an application and causes increased response times, reduced throughput, or degraded user experience under load conditions.

The purpose of bottleneck analysis is to identify areas of the application that require additional processing time or resources when handling user requests.

During the execution of the BlazeDemo load testing scenarios, no critical performance bottlenecks were observed. All requests were processed successfully and the application maintained stable performance throughout testing.

However, analysis of the generated Gatling reports revealed certain areas where response times increased as transaction complexity increased. These observations are discussed below.

12.2 Slowest Transactions

The slowest transactions observed during testing were associated with the Complete Booking workflow.

Among all executed scenarios, Scenario 3 generated the highest response times because it involved multiple business operations within a single transaction.

Workflow

Home Page



Find Flights



Choose Flight



Enter Passenger Details



Purchase Flight



Confirmation Page

Unlike Scenario 1 and Scenario 2, the Complete Booking scenario required:

- Additional form submission
- Passenger data processing
- Booking confirmation generation
- Multiple server-side operations

These additional processing steps resulted in higher response times compared to the simpler workflows.

Observation

The booking submission and confirmation generation requests required more processing time than flight search and flight selection requests.

Although these transactions were the slowest observed during testing, all requests completed successfully without failures.

12.3 Peak Response Times

Peak response time represents the highest response time recorded during execution.

Scenario 2 – Choose Flight

Metric	Value
Maximum Response Time	756 ms

The maximum response time remained below one second and did not indicate any performance concern.

Scenario 3 – Complete Booking

Metric	Value
Maximum Response Time	1620 ms

The highest response time observed during the entire project was 1620 milliseconds.

This peak occurred during the Complete Booking scenario.

Analysis

The increased response time can be attributed to:

- Additional HTTP requests
- Passenger data submission
- Booking processing activities
- Confirmation page generation

Even though a small number of requests required more processing time, the overall application remained responsive and stable.

Response Time Distribution

Category	Percentage
Less than 800 ms	90%
800 ms – 1200 ms	5%
Greater than 1200 ms	5%

The results show that the majority of requests completed quickly, while only a small percentage experienced higher response times.

12.4 Performance Constraints

Performance constraints are factors that can potentially affect application performance when user load increases.

During testing, the following observations were made:

1. Increased Processing During Booking Transactions

The Complete Booking workflow required more server-side processing compared to other scenarios.

Activities such as:

- Form validation
- Passenger data handling
- Purchase processing
- Confirmation generation

introduced additional processing overhead.

This resulted in slightly higher response times compared to the Search Flights and Choose Flight scenarios.

2. Response Time Growth with Workflow Complexity

As transaction complexity increased, response times also increased.

Comparison:

Scenario	Average Response Time
Choose Flight	476 ms
Complete Booking	516 ms

This indicates that more complex workflows naturally require additional processing time.

3. Limited Test Load Scope

The load tests were executed using:

- 20 Virtual Users
- 30-sSecond Ramp-Up

Because the user load was relatively moderate, the application did not reach its maximum capacity.

Higher loads such as:

- 50 Users
- 100 Users
- 200 Users

might reveal additional bottlenecks that were not visible during the current testing cycle.

4. External Network Dependency

The BlazeDemo application is hosted externally and accessed through the internet.

As a result, response times may be influenced by:

- Internet latency
- Network bandwidth
- Server availability
- External infrastructure performance

These factors can contribute to response time variations even when the application itself is functioning correctly.

12.5 Overall Bottleneck Assessment

Based on the execution results, no critical performance bottlenecks were identified in the BlazeDemo Flight Booking Application.

Key findings include:

- No failed requests were observed.
- Application stability remained consistent.
- 100% transaction success rate was achieved.
- Most requests completed within acceptable response time limits.
- Peak response times occurred only during the Complete Booking workflow.
- Response time increases were proportional to transaction complexity.

The observed behavior indicates that the application successfully handled the configured workload of 20 concurrent virtual users without significant performance degradation.

Therefore, the BlazeDemo application demonstrated good performance and stability under the tested load conditions, with only minor response time increases during complex booking transactions.